

Online Shoppers Purchasing Intention

Predicting Session-Level Conversion from In-Session Browsing Behaviour

IE 7275: Data Mining in Engineering

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1. Problem Setting

Retail businesses operate under finite operating and advertising budgets, and many smaller firms have moved to operate exclusively online in order to reduce the overhead costs associated with a physical storefront. Operating online has the advantage that raw behavioural data can be generated and collected far more easily, and in much larger quantities, than is possible in a brick-and-mortar setting. The trade-off, however, is that this abundance of data is considerably harder to interpret. In a physical store, an experienced operator or owner can simply read visual and contextual cues from a customer, such as body language, and form an immediate judgement about who is likely to make a purchase and who is merely browsing. On the other hand, during online shopping sessions, those cues are replaced by data represented by raw numbers that do not speak for themselves.

Without an effective way to glean insights and inferences from these data, smaller online retailers run the risk of misallocating their limited resources toward campaigns and site designs that do not convert visitors into paying customers. The cost of that misallocation falls disproportionately on small operators, for whom a wasted advertising cycle or an ineffective redesign represents a much more substantial share of an already constrained budget. The broader context for this project is therefore the conversion problem faced by a budget-constrained online retailer: how to turn an abundance of low-cost, hard-to-read session data into a small number of decisions about where to spend money and attention.

This project treats that question as a supervised learning problem. The central challenge is that purchase behaviour is rare relative to browsing behaviour, so the data are heavily imbalanced toward sessions that end without a sale. Any approach to the problem must contend with that imbalance directly, both in how models are trained and in how their performance is judged, because a method that is accurate on the common case while failing to identify buyers is of no practical use to a retailer.

2. Problem Definition

The primary objective of this project is to predict whether a browsing session will end in a purchase, using only the activity recorded during that session. The target of prediction is the

binary variable **Revenue**, which records whether a session generated a sale. This is framed as a binary classification task.

A secondary, equally important objective is interpretive rather than predictive: the ability to rank the available predictors by their contribution to purchase intent, so that the elements of the digital storefront and the behavioural patterns most strongly associated with conversion can be identified. The distinction matters because a retailer cannot act on a prediction alone; it needs to know which levers, among those it can actually control, are worth pulling. The project therefore aims both to flag high-intent sessions and to surface the actionable structure behind conversion.

The class imbalance is of particular importance, as across the cleaned dataset, 84.4% of sessions did not convert and only 15.6% did. A naive classifier that predicts “no purchase” for every session would therefore achieve 84.4% accuracy while identifying none of the buyers it was built to find. Accuracy is consequently a misleading measure of performance for this problem and is not treated as a priority. The project instead defines and evaluates success in terms of the model’s ability to recover the minority purchasing class, using metrics designed for imbalanced classification, principally the F1 score and the area under the precision–recall curve.

3. Data Source

The data used in this project are drawn from the Online Shoppers Purchasing Intention dataset, made available through the UCI Machine Learning Repository, and mirrored on Kaggle. The records were gathered from an anonymous real-world online retail site over a one-year period, with some features such as browser type anonymized as well. Several of the variables (specifically the bounce rates, exit rates, and page value figures) were captured through Google Analytics, while the remaining session counts and durations were recorded directly from site activity.

The dataset can be found via the following citation:

Sakar, C. (2018). *Online Shoppers Purchasing Intention Dataset*. UC Irvine, Machine Learning Repository. <https://doi.org/10.24432/c5f88q>

4. Data Description

The original dataset contains 12,330 records, each corresponding to a single browsing session collected over the one-year observation window. There are eighteen columns in total: seventeen predictors and one binary target, Revenue; the predictors comprise a mix of continuous measurements (durations, values), integer counts, and categorical descriptors of the session and the visitor. The data are complete, at least in the sense that no values are missing from any column.

The predictors fall into a few natural groups; a first group counts the number of pages of each type that a visitor viewed during the session (Administrative, Informational, ProductRelated), and a parallel group records the total time spent on each of those page types (Administrative_Duration, Informational_Duration, ProductRelated_Duration). A second group consists of engineered web-analytics metrics produced by Google Analytics: BounceRates and ExitRates, which describe how often visitors leave from a given page, and PageValues, which assigns a monetary value to the pages a visitor saw based on their proximity to past transactions. A SpecialDay indicator captures closeness to a notable shopping date, such as Valentine's Day, Mother's Day, or Black Friday. The remaining variables are categorical descriptors stored partly as text and partly as integer codes: the month of the visit, the visitor type, a weekend flag, and identifiers for the operating system, browser, geographic region, and traffic source.

Two features of the data require attention which will be elaborated upon later prior to use in modelling. The first is the presence of exact duplicate rows; the second is the heavy class imbalance in the target: purchasing sessions are far rarer than non-purchasing ones, which directly motivates the modelling choices described later. A sample of the variable names and roles are explained in Table 1, and the target distribution after cleaning is shown in Figure 1.

Table 1. Variables in the dataset and what they represent.

Variable	Type	Description
Administrative / Informational / ProductRelated	Integer	Number of pages of each type viewed in the session
*_Duration (three columns)	Continuous	Seconds spent on each corresponding page type
BounceRates	Continuous	Share of single-page sessions on a page (from Google Analytics)

Variable	Type	Description
ExitRates	Continuous	Share of sessions exiting from a page (from Google Analytics)
PageValues	Continuous	Average value of pages viewed, based on past transactions
SpecialDay	Continuous	Proximity of the session to a special shopping day, i.e. holidays
Month / VisitorType / Weekend	Categorical	Calendar month, new vs. returning visitor, weekend indicator
OperatingSystems / Browser / Region / TrafficType	Categorical	Coded technical and source descriptors of the session
Revenue	Binary target	Whether or not the session generated a purchase

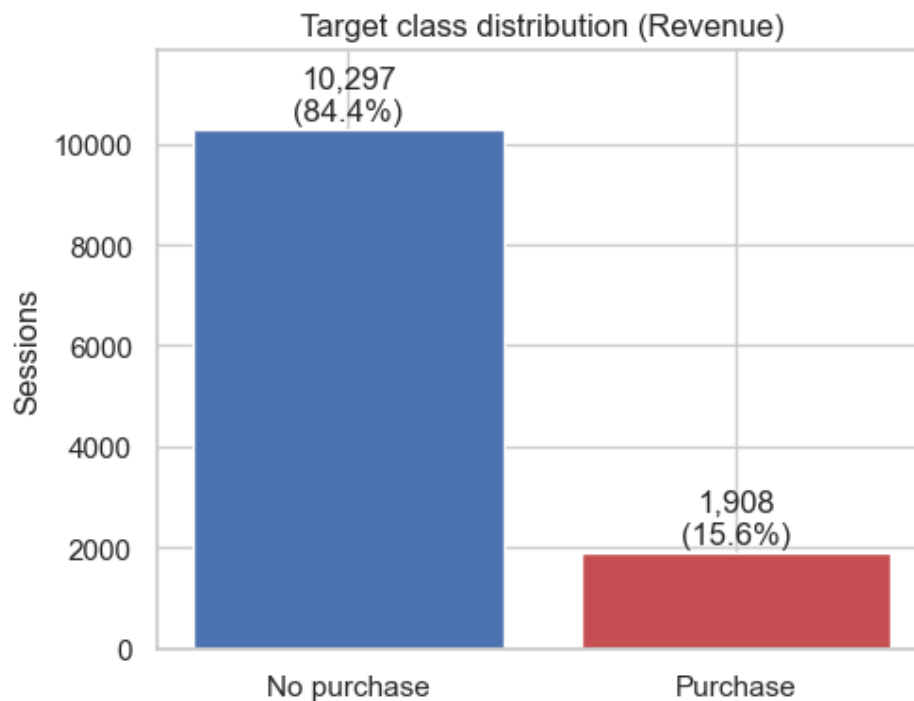


Figure 1. Target class distribution after de-duplication. Only 15.6% of sessions convert into sales.

5. Data Exploration

Exploratory data analysis served to both verify the integrity of the data and to check for underlying structures to guide both feature handling and the interpretation of results. The exploration combined summary statistics with a set of targeted visualizations of behaviour, correlation, and conversion across segments.

As mentioned previously, a quick check for completeness confirmed that the dataset contained no missing values, as reported by the creators. A check for exact duplicate rows, however, identified 125 identical records. Duplicates are a specific hazard in this context because an identical session appearing in both the training and test partitions would leak information into the held-out performance estimate, thereby inflating it artificially. These 125 duplicate rows were therefore removed, leaving 12,205 unique sessions. Notably, all of the removed duplicates belonged to the non-purchasing class, so the count of purchasing sessions was unchanged at 1,908; the cleaned data are comprised of 84.4% non-purchasing entries and 15.6% purchasing entries.

Summary statistics on the duration variables revealed an extreme right tail, which makes sense; time cannot go negative and thus a long left tail is an impossibility, while people frequently step away and idle while leaving pages open on their devices. Browsing time is recorded in seconds, and at least one session registered nearly eighteen hours of continuous product-page viewing, which is extremely unlikely to be active shopping. Two kinds of unusual sessions were examined, namely, ultra-short sessions with browsing times of 0, and ultra-long browsing times belonging to the long right tail. The ultra-short sessions that recorded page visits but zero browsing time, of which there were 589 (428 of them single-page) and behaved as immediate bounces with a conversion rate of only 0.51%, were treated as natural behaviour and retained. The long-duration tail was examined using Tukey's rule, flagging sessions whose duration-per-page exceeded the third quartile by more than three times the interquartile range. This identified 285 sessions, or 2.34% of the data. Rather than converting at an elevated rate, as genuinely engaged shoppers would, these long sessions converted at only 10.9%, below the overall rate of 15.6%, which indicates idle time rather than active shopping.

Despite this, all unique rows were retained for modelling. The tree-based models used in the study are robust to outliers, and the logistic models are protected by standardization and shrinkage. Moreover, the lower conversion rate among very long sessions is in and of itself an informative signal rather than noise to be removed.

A comparison between converting and non-converting sessions shows a clear behavioural difference. Converting sessions spend more time viewing product-related pages, as expected, but the distinction is sharper than a simple count of pages would suggest: buyers not only visit more

pages, but also spend more time per page. It's thus likely that it's the depth of engagement rather than the sheer volume of browsing that separates buyers from non-buyers. Figure 2 contrasts the two groups on product-related browsing time and on duration per page.

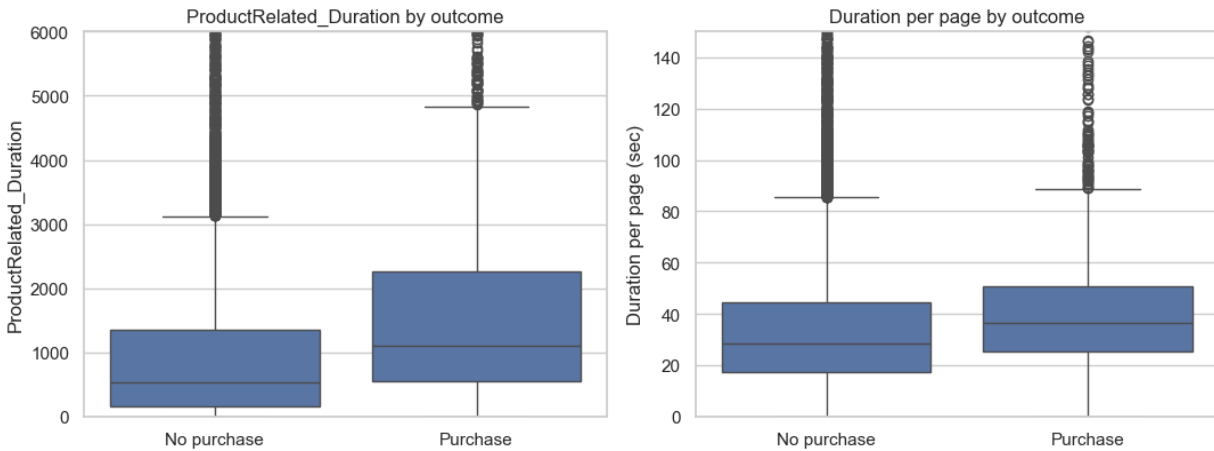


Figure 2. Product-related browsing time (left) and duration per page (right), by session outcome.

A correlation matrix of the numeric features (Figure 3) shows several strong, unsurprising relationships. The number of product-related pages and the engineered total page count are almost perfectly correlated (0.997), as are product-related and total duration (0.994). Bounce rates and exit rates are also strongly related (0.902). Among all predictors, PageValues is the variable most correlated with the target, for good reason, which will be elaborated upon later.

These correlations are structural rather than coincidental, as total counts are sums that include their components, and bounce and exit rates measure closely related leaving behaviour. Because the regularized logistic models absorb collinearity through shrinkage and the tree models are unaffected by it, no dimensionality reduction was applied. Principal component analysis in particular was avoided because it would obscure the per-feature interpretability that the secondary objective requires. The most strongly correlated pairs are listed in Table 2.

Table 2. The most strongly correlated numeric feature pairs.

Feature pair	Absolute correlation
ProductRelated - TotalPages	0.997
ProductRelated_Duration - TotalDuration	0.994
BounceRates - ExitRates	0.902
ProductRelated - ProductRelated_Duration	0.860
ProductRelated_Duration - TotalPages	0.860

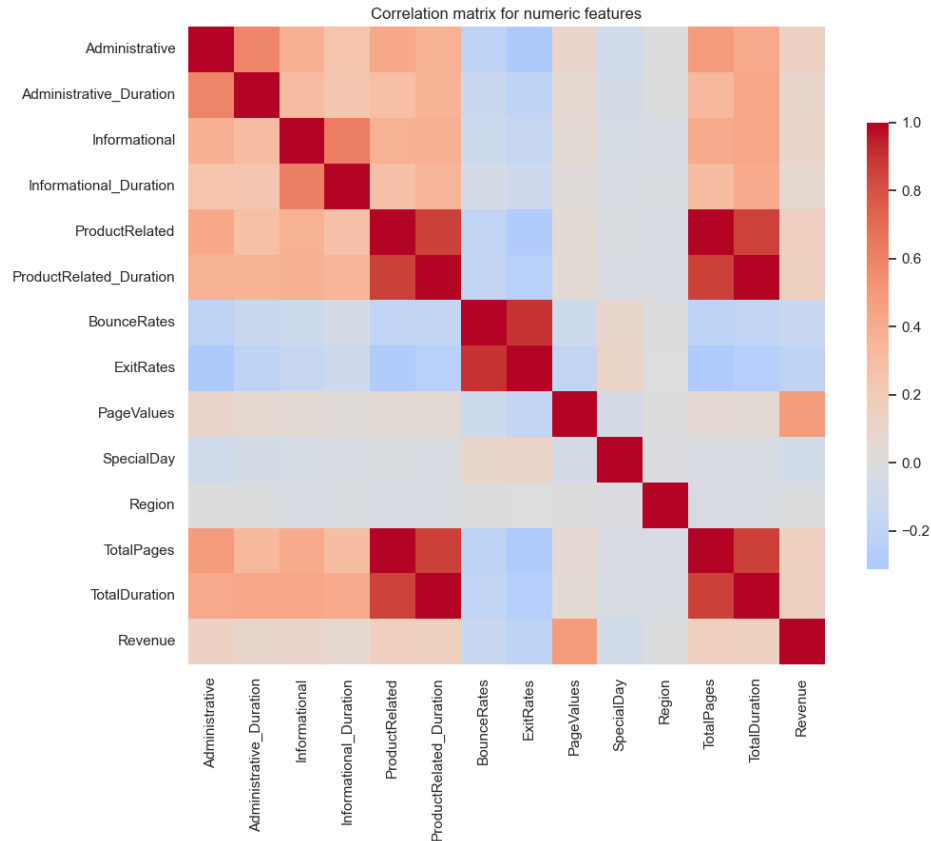


Figure 3. Correlation matrix for the numeric features.

Conversion rates were then examined across calendar, visitor, and engagement segments, and a strong seasonal pattern was observed, with an autumnal run-up from August through November. (Figure 4). No sessions were recorded in January or April. Conversion is lowest in February at about 2% and rises steadily through the year, peaking in November at roughly 25%, more than double the spring rate, before falling back in December.

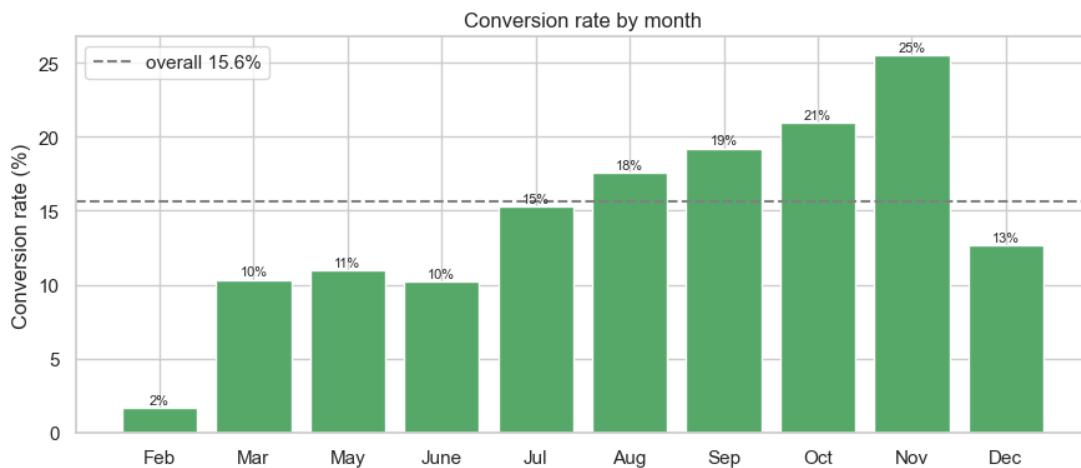


Figure 4. Conversion rate by month, against the overall 15.6% average.

Visitor type and the weekend flag show smaller but consistent effects, though formal hypothesis testing was not conducted to check for statistical significance (Figure 5). New visitors convert at about 24.9%, nearly double the 14.1% rate of returning visitors, with the small “other” group in between. Weekend shoppers convert at 17.5% against 15.1% for weekday shoppers, an even more minor difference.

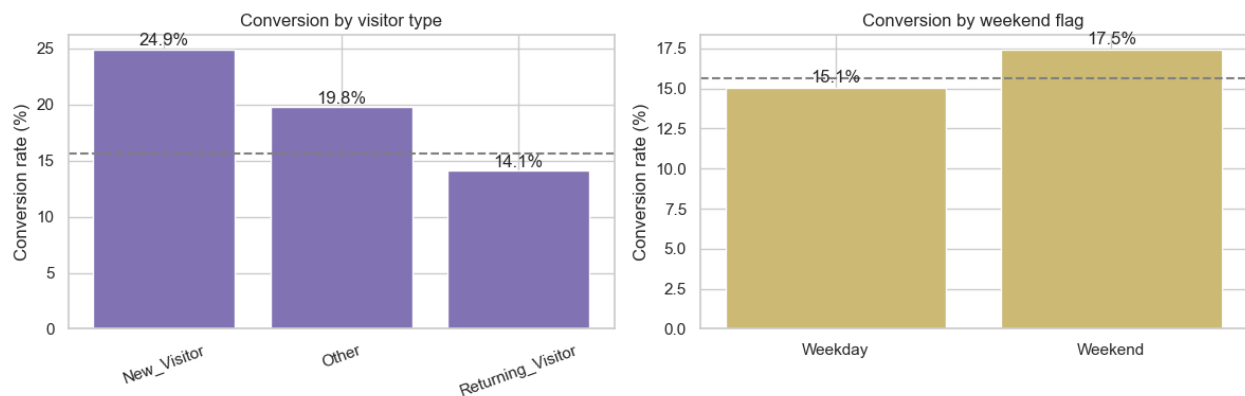


Figure 5. Conversion by visitor type (left) and by weekend flag (right).

The strongest single distinction in the data concerns PageValues. About 22.4% of sessions reach at least one value-bearing page, and those sessions convert at 56.3%. Sessions that never reach a value-bearing page, with PageValues equal to zero, convert at only 3.9% (Figure 6). This metric has the most separating power in the dataset. Its derived nature, however, has important consequences for how it can be used, a point revisited in the results: because it is calculated by Google Analytics from past transactions, a high page value largely reports that a session is moving closer to a purchase, and does not describe something the retailer can directly change.

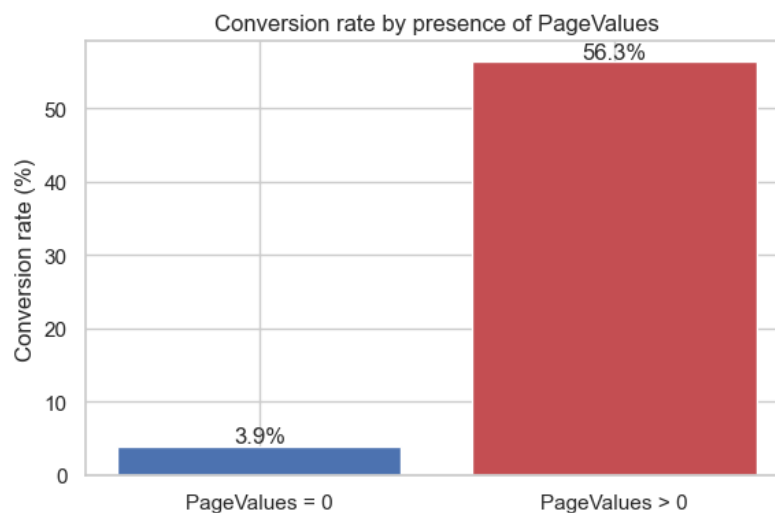


Figure 6. Conversion rate by presence of a non-zero PageValues.

6. Data Mining

The primary task was supervised binary classification, optimized for the recovery of the minority purchasing class. Several preparatory transformations were carried out before any model was fitted.

The first was feature engineering and transformation. Two aggregate features, TotalPages and TotalDuration, were created by summing the page-count and duration columns; these supported the per-page duration analysis used in outlier screening and were carried into modelling. The categorical descriptors were one-hot encoded so that no artificial ordering was imposed on codes such as browser or region. This expanded the predictor set from nineteen columns (seventeen, plus our two engineered features) to seventy-five. Two parallel views of the data were prepared so that each model family could be fed appropriately: a standardised view, in which the numeric features were centred and scaled, for the logistic models, and a raw view in which numeric features were left unscaled for the tree-based models, which do not benefit from scaling.

Care was also taken regarding the train–test split. The data were divided 80/20 into training and test partitions using a fixed random seed for reproducibility. The split was stratified on the target so that both partitions preserved the 15.6% positive rate of the full data, yielding a training set of 9,764 sessions and a test set of 2,441 sessions, each with the same class balance.

The class imbalance was addressed differently across model families, with care taken to avoid information leakage. For the baseline logistic regression, which has no built-in mechanism for imbalance, the positive class in the training set was randomly oversampled with replacement until the training data were balanced; the test set was left untouched. For every other model, the imbalance was handled inside cross-validation by passing class weights, using `class_weight="balanced"` for the LASSO and random forest models and `scale_pos_weight` set to the negative-to-positive ratio (approximately 5.4) for the gradient-boosted model. Using reweighting inside cross-validation rather than resampling beforehand keeps the resampling out of the validation folds and prevents leakage across folds.

Hyperparameters were tuned using ten-fold stratified cross-validation with a single fixed set of folds shared across all grid searches, so that every model was evaluated on identical partitions. Cross-validation was scored on the F1 measure to focus on the minority class.

7. Models and Methods

Six classifiers were trained and compared, spanning interpretable linear models and higher-capacity tree ensembles, together with an ensemble that combines them. The models are as follows:

Logistic regression (baseline). An unregularized logistic regression fitted on the oversampled training set, included as an easily interpretable reference point against which the other methods are measured.

LASSO logistic regression (lambda.min). An L1-penalised logistic regression whose regularization strength was tuned over fifty candidate values, selecting the value that minimized cross-validated log loss. The L1 penalty performs embedded feature selection, driving the coefficients of uninformative predictors to zero. This configuration retained 34 features.

LASSO logistic regression (lambda.1se). The most parsimonious model whose cross-validated error remained within one standard error of the best, following the one-standard-error rule familiar from R's `glmnet` package. This model retained only 12 features in comparison. The rule and its implementation are described in further detail below but align with Hastie's "one standard error" rule as described on pages 61 and 244 of *The Elements of Statistical Learning* (Hastie et al., 2009).

Random forest. A bagged ensemble of decision trees fitted on the raw feature view, with the number of trees and the maximum tree depth tuned by cross-validation. The selected configuration used 300 trees at a maximum depth of 12.

XGBoost. A gradient-boosted tree ensemble, well suited to imbalanced data, with the positive class up-weighted through the `scale-pos-weight` parameter. Tuning over the number of trees, depth, and learning rate selected 1,500 trees at depth 5 with a learning rate of 0.02 and row subsampling of 0.9.

Soft-vote ensemble. An ensemble that averages the predicted probabilities of the LASSO, random forest, and XGBoost models, included to test whether combining linear and tree-based learners would yield any advantage over the strongest individual model.

Elaborating upon the two LASSO models, they differ only in how the strength of the L1 penalty is chosen, and that choice follows the one-standard-error rule familiar from Hastie's description,

or the implementation in the R `glmnet` package. For each candidate penalty, ten-fold cross-validation produces an estimate of out-of-sample error, here measured as log loss (equivalently, binomial deviance), together with the standard error of that estimate across the ten folds. The `lambda.min` rule selects the penalty that minimizes the mean cross-validated deviance. The one standard error rule is deliberately more conservative: among all penalties whose mean deviance lies within one standard error of that minimum, it selects the strongest penalty, that is, the largest value of `lambda`. The result is the most parsimonious model whose cross-validated performance is practically statistically indistinguishable from the best.

Applied to this dataset, the penalty minimizing cross-validated deviance corresponded to an inverse-strength parameter of $C = 0.16$, with a mean deviance of 0.441 and a standard error of 0.011 across the folds. The one-standard-error threshold is therefore $0.441 + 0.011 = 0.452$, and the most strongly regularized penalty whose deviance stays within that band corresponded to $C = 0.0139$. Because the inverse-strength parameter moves opposite to the penalty, the smallest C inside the band represents the largest penalty (or `lambda`), consistent with the rule. Tightening the penalty in this way reduced the number of retained features from 34 under `lambda.min` to 12 under `lambda.1se`, and, as the performance table later shows, slightly improved test-set F1 rather than degrading it, indicating that the discarded features were adding variance rather than signal.

As the one-standard-error rule is not implemented in the modelling library used, it was applied directly to the cross-validation results: the standard error of each penalty's mean score was computed as the across-fold standard deviation divided by the square root of the number of folds, the one-standard-error band was formed around the best score, and the largest qualifying penalty was selected and used to refit another LASSO model on the full training set.

8. Performance Evaluation

Because of the class imbalance, the models were judged primarily on their ability to recover the minority purchasing class rather than on overall accuracy. The reported metrics are the F1 score, precision, and recall at the default 0.5 decision threshold, together with the area under the precision–recall curve (PR-AUC) and the area under the ROC curve (ROC-AUC), both of which assess ranking quality across all thresholds. The F1 score and PR-AUC are the most informative measures for this problem, since both reflect performance on the rare positive class. The PR-

AUC was plotted against the positive class rate as a baseline, and the ROC-AUC was plotted with the identity function in mind as a baseline. All metrics were computed on the held-out test set of 2,441 sessions.

Table 3 ranks the six models by F1 score. XGBoost is the strongest model on every metric, with an F1 of 0.693 and a PR-AUC of 0.735. The soft-vote ensemble and the random forest follow, and the three logistic models trail behind somewhat. The more compact twelve-feature LASSO outperforms both the unregularised baseline and the fuller lambda.min LASSO on F1, indicating that the additional parsimony improved generalisation rather than harming it. It must be noted that the soft-vote ensemble was likely carried by the performance of XGBoost, which in other terms means that it was likely brought down by the comparatively poorer performance of the Random Forest and LASSO models. As such, as a model, it is not of much interest to us for gleaning results.

Table 3. Test-set performance, ranked by F1 score ($n = 2,441$).

Model	F1	Precision	Recall	PR-AUC	ROC-AUC
XGBoost	0.693	0.618	0.791	0.735	0.927
Soft-vote ensemble	0.673	0.591	0.780	0.726	0.926
Random forest	0.669	0.588	0.775	0.688	0.920
LASSO (lambda.1se)	0.627	0.537	0.751	0.649	0.901
LASSO (lambda.min)	0.608	0.503	0.767	0.646	0.900
Logistic (baseline)	0.602	0.495	0.770	0.644	0.899

The ranking and precision–recall curves in Figure 7 make the comparison concrete across thresholds. The separation among models is much clearer in the precision–recall view, which is the more relevant view under such a strong class imbalance, than in the ROC view. XGBoost leads on PR-AUC by 0.091 over the baseline, whereas the spread in ROC-AUC is only 0.028, confirming that the gains from the stronger model are indeed concentrated where they matter most for a low-frequency positive class.

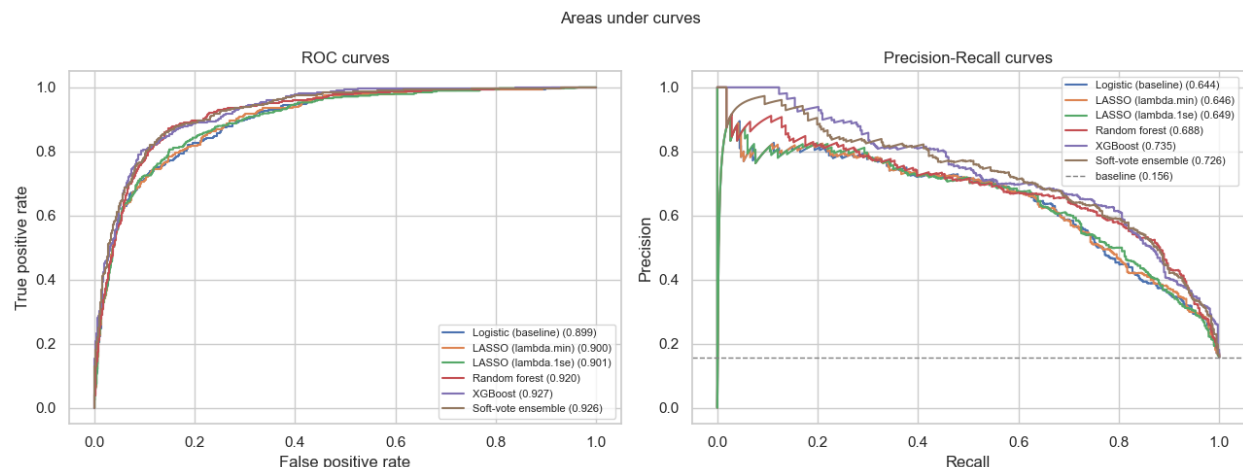


Figure 7. ROC curves (left) and precision–recall curves (right) for all six models.

The confusion matrices at the 0.5 threshold show the precision–recall trade-off directly. Figure 8 reports the raw counts against the 2,441-session test set, and Figure 9 row-normalises them so that the per-class rates can be compared across models. XGBoost correctly classifies 90.9% of non-purchasing sessions and recovers 79.1% of true purchases, the best recall of any model; the logistic models recover a comparable share of buyers but at the cost of more false positives among non-buyers. The default threshold favours recall, which is appropriate when missing a buyer is more costly than showing an unnecessary offer, and it can be tuned more carefully to a retailer’s specific costs, given additional contextual information such as advertising spend, profit per sale, and more.

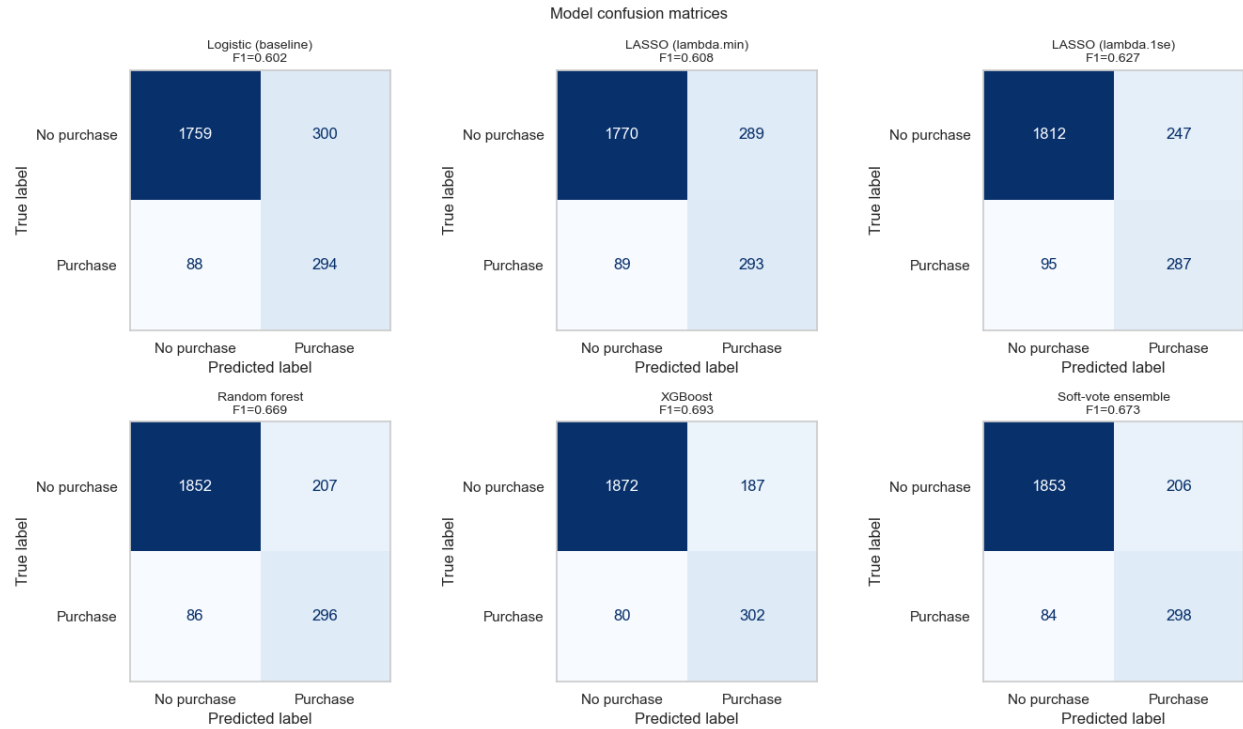


Figure 8. Confusion matrices in raw counts at the 0.5 threshold (test set, $n = 2,441$).

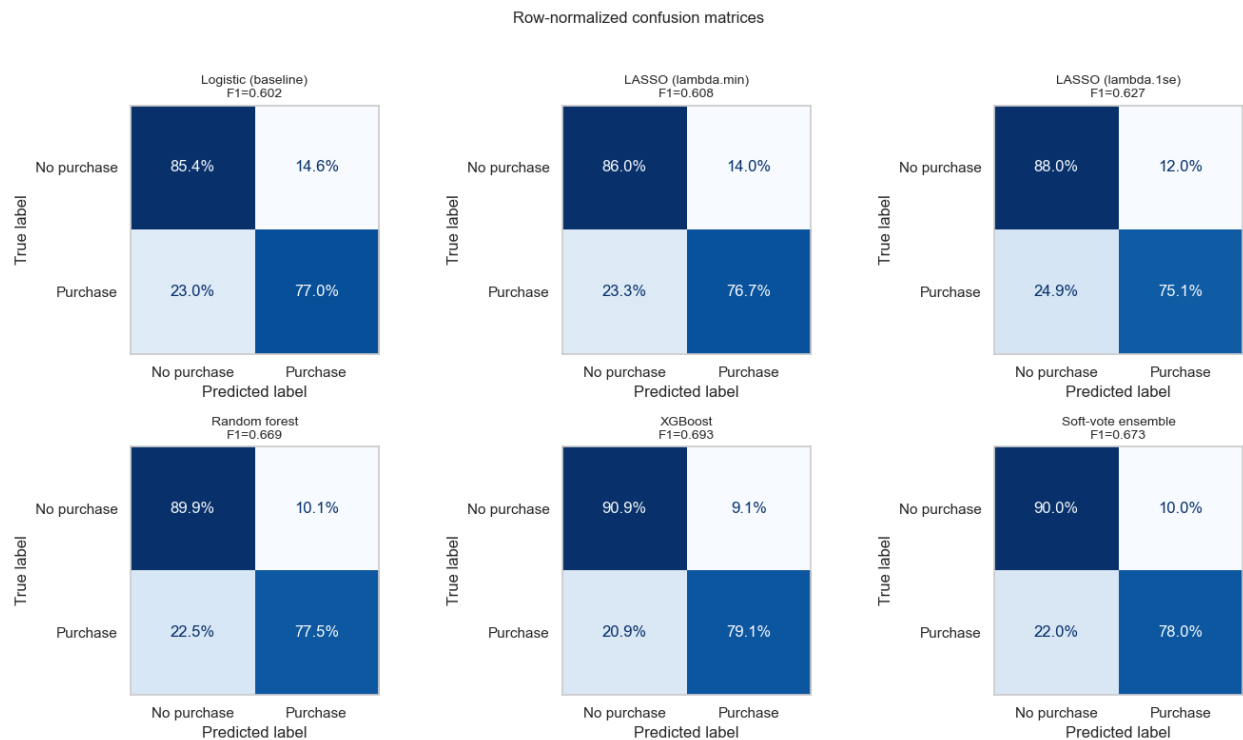


Figure 9. Row-normalised confusion matrices at the 0.5 threshold.

9. Project Results

Three of the models, the parsimonious LASSO using λ_{1se} , the random forest, and XGBoost, expose a measure of feature contribution, whether through standardised coefficients in the case of LASSO or through split-based importance from our tree-based models. These were examined to address the project's secondary objective of ranking predictors by their contribution to purchase intent. The findings are consistent across all three (Figures 10 and 11).

PageValues was found to dominate in every model by a wide margin. In the standardised LASSO, its coefficient (about 2.05) dwarfs all others, and it ranks first in the tree-based importance for both Random Forest and XGBoost as well. This is consistent with the exploratory finding that value-bearing sessions convert at roughly 56% against only ~4% otherwise. Again, however, it must be kept in mind that PageValues is a Google Analytics metric built from past transactions: a high value largely indicates that a session has already moved close to a purchase. It is therefore an excellent detector of high-intent sessions while they are live, but it is not a variable that the store can adjust to manufacture conversions.

Exit and bounce rates are the most actionable signals. Exit rate carries the largest negative coefficient in the LASSO model (about -0.48), and both exit and bounce rates rank highly across models. Unlike PageValues, these metrics point to where visitors are leaving the site, which is precisely where changes to the storefront itself can be targeted.

Seasonality and visitor type sharpen targeting. November is a strong positive signal in both the LASSO and the gradient-boosted model, reflecting the autumn conversion peak, while May carries a negative weight. Visitor type also contributes: returning visitors carry a negative coefficient relative to new visitors, who convert at nearly twice the rate. The most plausible reading is that new visitors often arrive through high-intent channels such as search results or targeted advertisements, so the result argues for prioritising new-visitor sessions on arrival rather than simply buying more traffic.

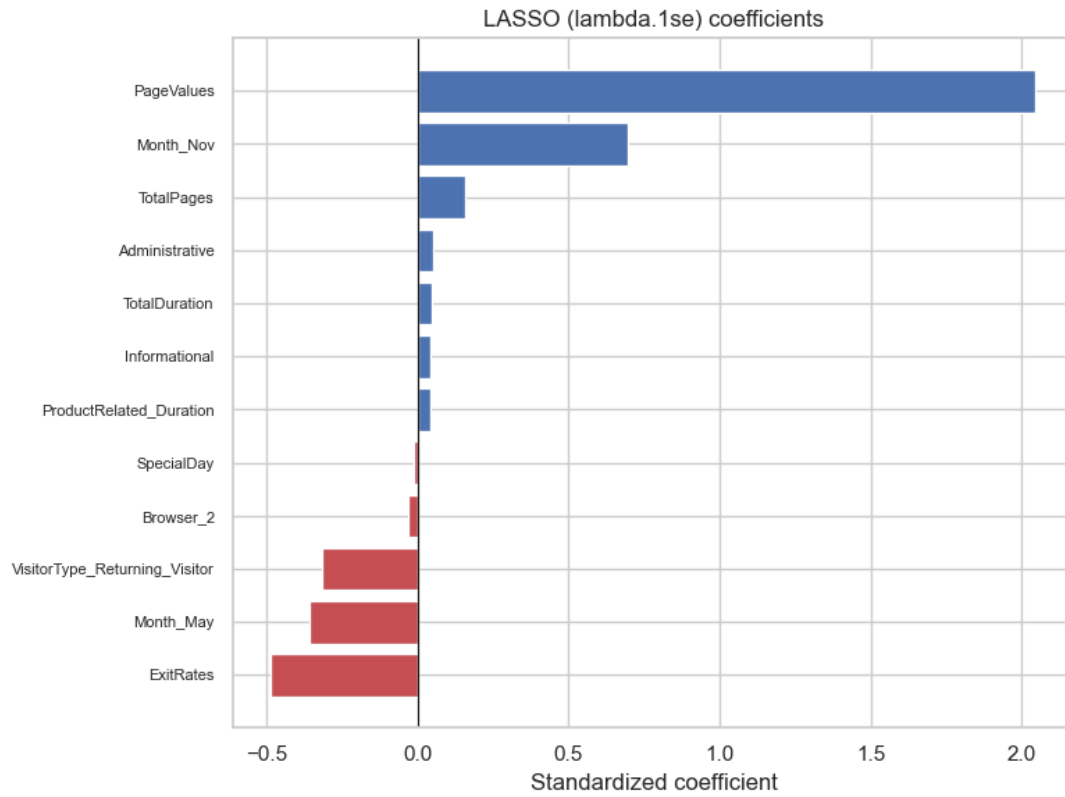


Figure 10. Standardised coefficients of the twelve-feature LASSO (lambda.1se) model.

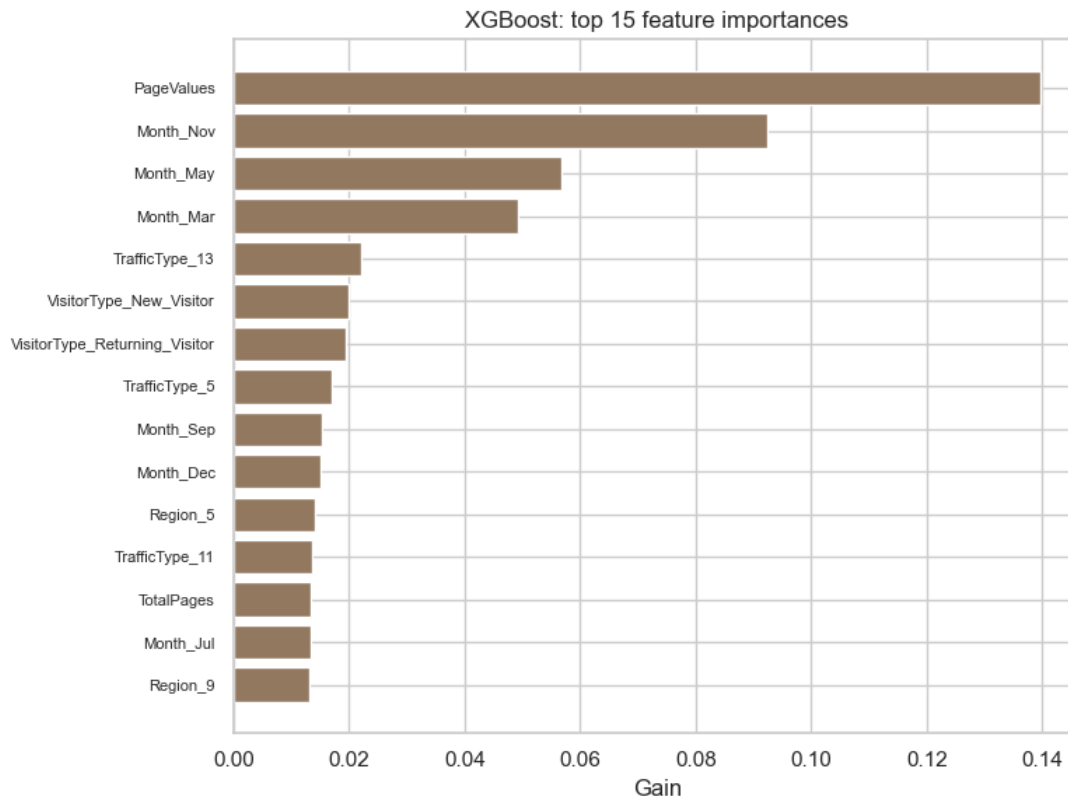


Figure 11. Top fifteen feature importances (gain) from the XGBoost model.

On the modelling side, XGBoost is the clear leader, with the soft-vote ensemble close behind. The ensemble, however, offers no practical advantage as mentioned above and is set aside. The overall gap between the best tree model and the linear models is modest: the twelve-feature LASSO already beats the unregularised baseline, which means a compact and readable model recovers most of the available performance. Interpretability therefore comes at relatively little cost, but the XGBoost model can be retained for making predictions while using the coefficients from the LASSO model to guide adjustments to the storefront as necessary.

10. Impact of the Project Outcomes

Practically, such exploration and modeling enables a small online retailer to pursue an evidence-backed direction for allocating a limited advertising and design budget, derived entirely from recorded in-session behaviour. The predictive models and the feature rankings translate into a small set of concrete recommendations:

- Page value ought to be used for real-time signaling rather than storefront redevelopment. The page-value signal can flag high-intent sessions while they are live, for example to trigger a purchase incentive such as a coupon for items being viewed.
- Use exit and bounce rates to locate the pages visitors most often leave from, as this is the most directly actionable point for improving retention through changes to the storefront.
- Concentrate promotional effort on the autumn run-up to December, when conversion rises steadily from August through its November peak, rather than spreading the budget evenly across the year.
- Prioritise new visitors on first arrival, which can be detected via cookies, fingerprints, or trackers, because new visitors convert at nearly twice the rate of returning ones. One of the most efficient actions is simply to make the most of new-visitor sessions when they appear, rather than spending more on advertising to chase additional traffic.

The models used support these recommendations in multiple regards, both operationally and descriptively. At the default threshold the best model already recovers close to four-fifths of all buyers, and the threshold can be tuned to a retailer's own economics, balancing the cost of a missed sale against that of a wasted offer. Interpretability is preserved on two fronts: the parsimonious twelve-feature LASSO offers a transparent linear account of which factors matter

via its coefficients, while the tree-based importances provide an explainable narrative that a non-technical owner or operator can follow by simply going down the list of feature importances. Because the underlying drivers are all general features of online shopping behaviour, the findings may be transferable to other online storefronts, subject to store similarity and local validation.

A few limitations affected these conclusions and point toward the extensions below. First, the reported performance rests on a single stratified train/test split, so the figures are best read as point estimates; repeating the evaluation across several splits, or reporting the variation already available across the cross-validation folds, would help provide a measure of uncertainty for the model ranking and confirm that the narrow gaps among the closely grouped models are meaningful. Hypothesis testing could be applied during the EDA to ensure that minor differences that were caught by eye between groups were in fact statistically significant and not merely noise. Second, several predictors are near-duplicates of one another, with the engineered totals almost perfectly correlated with their component columns; this does not harm predictive performance, but it does allow a penalised model to distribute weight somewhat arbitrarily among collinear features, so the magnitude of any single small coefficient is best read with that in mind. Third, because the models were trained with class reweighting and oversampling, their predicted probabilities are tuned for ranking rather than calibrated to true purchase rates, so using them directly for live offer targeting, as the recommended, would first require some degree of recalibration. Ultimately, though, none of these caveats change the main findings, as each simply marks the boundary of what this single-split, session-level study can safely state.

Several extensions would further strengthen and operationalize the results, but would require more knowledge of the storefront and direct access to the environment the data were gathered from. A/B testing of storefront changes guided by high-exit pages would move the analysis from association toward causality. A formal business analysis of profit margins, break-even points, costs, and more would tie the decision boundary more explicitly to revenue or profit per conversion alongside the cost of intervention. Calibration and ongoing monitoring of the predicted probabilities would be required before live deployment, for instance in real-time offer targeting. Finally, collecting richer data, such as more granular clickstream sequences, session recency, and channel-level detail, would enable the development more refined models than mere session-level aggregates allow.

11. Conclusion

Ultimately, in-session browsing behaviour is still able to predict purchase conversion reasonably well despite a heavy class imbalance, with the gradient-boosted tree-based model (XGBoost) reaching an F1 of 0.693 and a PR-AUC of 0.735 on held-out data. Even a compact twelve-feature logistic model (LASSO with $\lambda = 1\text{se}$) still recovers most of that performance, while improving interpretability by yielding coefficients for predictors. Beyond prediction, the analysis still enables the store operator to make decisions and act on both strengths and weaknesses alike: while page value detects intent but cannot be steered, exit and bounce rates indicate where the store can intervene, and seasonality and visitor type help guide the timing and targeting of that effort. Together, these findings turn a relatively abstract, hard-to-read, high-dimensional stream of session data into a small, digestible set of recommendations for a budget-constrained online small business.

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